

MA1521 Calculus for Computing

Lecture Group 2

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Lesson 04: Friday 21 August 2026

(Week 2)

Remark. Q03.7 to Q03.9 are leftover exercises from the Tuesday class in Week 2.

Remark. Q03.7 and Q03.8 are on Section 9.9 of Thomas' Calculus.

Q03.9-Q04.10 are on topics in Sections 11.1-11.4 of Thomas' Calculus.

Q04.11-Q04.15 are on topics in Section 11.5.

Q03.7. (i) Let $f : I \rightarrow \mathbb{R}$ be a function defined on an interval I , and take a point $a \in I$. Let $g(x) = b + c(x - a) + d(x - a)^2$ be a quadratic polynomial, where b, c, d are real constants. Suppose

$$g(a) = f(a), \quad g'(a) = f'(a) \quad \text{and} \quad g''(a) = f''(a).$$

Find b, c, d in terms of $f(a)$, $f'(a)$ and $f''(a)$.

(ii) Now let $a = 1$ and suppose $f(1) = 2$, $f'(1) = 3$ and $f''(1) = 5$. Write down $g(x)$ in this case. Then using $g(x)$ as an approximation of $f(x)$, find an approximate value of $f(1.05)$.

Q03.8. Using the Maclaurin series of e^x given by

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots, \quad -\infty < x < \infty,$$

find the Taylor series of the function $f(x) = e^{2x-2}$ at $x = 3$.

Q03.9. Consider the vectors $\vec{A} = \langle 1, 5, 6 \rangle$ and $\vec{B} = \langle 2, 1, -3 \rangle$.

(i) Find $2\vec{A} - \vec{B}$.

(ii) Find the lengths of $\vec{A}$ and $\vec{B}$.

(iii) Find the unit vector with the same direction as $\vec{B}$.

(iv) Find the dot product $\vec{A} \cdot \vec{B}$.

(v) Hence find the angle between $\vec{A}$ and $\vec{B}$. Give your answer in radians, correct to two decimal places.

Q04.1. Consider the vectors $\vec{A} = \langle 5, 6 \rangle$ and $\vec{B} = \langle 1, -3 \rangle$ in the xy -plane.

(i) Find $\vec{A} + 3\vec{B}$.

(ii) Find the unit vector with the same direction as $\vec{B}$.

(iii) Find the angle between $\vec{A}$ and $\vec{B}$. Give your answer in degrees, correct to two decimal places.

Q04.2. Consider the unit vector $\mathbf{u} = \frac{1}{\sqrt{6}}\mathbf{i} - \frac{1}{\sqrt{6}}\mathbf{j} + \frac{2}{\sqrt{6}}\mathbf{k}$ and the vector $\vec{A} = 3\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$. Find the component of $\vec{A}$ along the direction of $\mathbf{u}$.

Q04.3. Consider the two vectors $\vec{A} = \langle 0, -2, -2 \rangle$ and $\vec{B} = \langle 1, 1, -1 \rangle$. Are $\vec{A}$ and $\vec{B}$ orthogonal to each other? Justify your answer.

Q04.4. Consider the vectors $\vec{A} = 6\mathbf{j} + 6\mathbf{k}$ and $\vec{B} = 2\mathbf{i} - \mathbf{j} - \mathbf{k}$. Write $\vec{A} = \vec{C} + \vec{D}$ as a sum of two vectors such that $\vec{C}$ is parallel to $\vec{B}$, and $\vec{D}$ is perpendicular to $\vec{B}$.

[Suggestion: First write $\vec{C} = \alpha \vec{B}$ so that $\vec{A} = \alpha \vec{B} + \vec{D}$, where α is a real number. Determine α by using the dot product. Then proceed to get $\vec{C}$ and $\vec{D}$.]

Q04.5. Consider the vectors $\vec{A} = \langle 5, 6 \rangle$ and $\vec{B} = \langle 1, -3 \rangle$ in the xy -plane. Find $\det(\vec{A}, \vec{B})$ and $\det(\vec{B}, \vec{A})$.

Q04.6. Consider the vectors $\vec{A} = \langle 1, 2, 3 \rangle$, $\vec{B} = \langle -1, -1, 2 \rangle$ and $\vec{C} = \langle 0, 1, -1 \rangle$ in space. Find $\det(\vec{A}, \vec{B}, \vec{C})$.

Q04.7. Consider the vectors $\vec{A} = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\vec{B} = \mathbf{j} - \mathbf{k}$. Find the cross product $\vec{A} \times \vec{B}$.

Q04.8. Consider the points $\mathbf{A} = (5, 2, 0)$, $\mathbf{B} = (2, 6, 1)$, $\mathbf{C} = (2, 4, 7)$ in space. Find the area of the parallelogram with two adjacent sides given by the line segments $\mathbf{AB}$ and $\mathbf{AC}$.

[Suggestion: First write down the vectors $\overrightarrow{\mathbf{AB}}$ and $\overrightarrow{\mathbf{AC}}$. Here $\overrightarrow{\mathbf{AB}}$ denotes the vector with initial point at $\mathbf{A}$ and terminal point at $\mathbf{B}$, etc.]

Q04.9. Consider the vectors $\overrightarrow{\mathbf{A}} = \mathbf{i} + 3\mathbf{k}$, $\overrightarrow{\mathbf{B}} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\overrightarrow{\mathbf{C}} = 5\mathbf{i} + 4\mathbf{k}$ in space. Find the volume of the parallelepiped spanned by the above three vectors.

Q04.10. Suppose ***A***, ***B***, ***C*** and ***D*** are four points in space such that $\det(\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}) = 0$. Is it true that the 4 points are coplanar (i.e. they all lie on a plane in space)? (You do not need to justify your answer.)

Q04.11. Find an equation of the plane containing the point $(6, 3, 2)$ and perpendicular to the vector $-2\mathbf{i} + 5\mathbf{j} + \mathbf{k}$.

Q04.12. Consider the plane with equation $x + 3y - 7z - 8 = 0$. Find a normal vector of the plane.

Q04.13. Consider the two following two planes:

$$P_1 : x + 3y - 7z - 8 = 0,$$

$$P_2 : -2x - 6y + 14z = 5.$$

Are the two planes P_1 and P_2 parallel? Justify your answer.

Q04.14. Find an equation of the plane passing through the points $\mathbf{A}(3, 3, 0)$, $\mathbf{B}(3, 0, 1)$ and $\mathbf{C}(0, 2, 1)$.

Suggestion: First find two vectors parallel to the plane, and then find a vector normal to the plane.

Q04.15. Find an equation of the plane P_1 that passes through the points $\mathbf{A} = (-1, 1, -1)$ and $\mathbf{B} = (2, 2, 1)$, and is perpendicular to the plane $P_2 : 4x - 6y + 2z = 1$.

Suggestion: First find two vectors parallel to the plane P_1 .